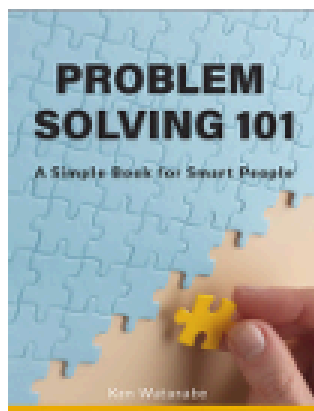




PROBLEM SOLVING 101

A Simple Book for Smart People

Ken Watanabe



The Big “So What”

Everyone faces problems, be it students, working professionals, or a country's prime minister. In this book, Watanabe presents a simple but powerful approach that can be used by anyone to solve any problem, big or small.



Introduction



Prior to writing this book, Watanabe worked as a consultant with McKinsey & Company, helping companies worldwide to solve complex problems. This book captures the problem-solving tools and approach that he used.

- He originally wrote *Problem Solving 101* as a guide to critical-thinking for Japanese schoolchildren. The book was so effective that it quickly became an international bestseller for readers of all ages.



This method and approach in this book can guide and expand your thinking, to help you find solutions to any problem, dream or goal—be it to learn a new skill, develop your dream career, or solve a global crisis.

- In this summary, we'll be presenting the key ideas in 2 parts:
 - (i) Mindset and approach for effective problem-solving; and
 - (ii) The 4 steps and toolkit for solving any problems. We'll also explore 2 detailed examples to illustrate how to apply the problem-solving tools and techniques.

KEY QUOTES

“There's a fundamental approach to solving these real-life problems, one that can consistently lead you to effective and satisfying solutions.”

“Being a problem solver isn't just an ability; it's a whole mind-set, one that drives people to bring out the best in themselves and to shape the world in a positive way.”



Part 1: Becoming a Problem-Solving Kid



The Problem-Solving Mindset



Problem-solving isn't an inborn talent; it's a set of skills, attitudes, and processes that can be learned.



Watanabe uses 5 types of problem-solvers to illustrate the ingredients behind great problem-solving:

- Defeatists give up easily and feel victimized by circumstances. They choose to avoid problems, and give up when confronted by the slightest difficulties. They say things like "I can never do that," and blame others for not understanding or supporting them.
- Critics are always shooting down others' plans, without lifting a finger to solve the problem themselves. They stay on the sidelines, complaining, criticizing, and saying things like "That's a stupid idea" or "I told you so."
- Daydreamers are always dreaming up new ideas. But, they keep their heads in the clouds, refusing to explore tangible plans or take concrete action. They can be heard saying things like: "If only this could happen" or "Don't bother me with details!"
- Dynamos jump into action without planning ahead. They're enthusiastic, optimistic, and perseverant. But, they don't stop to think things through, or to understand a problem before charging ahead. As a result, they waste time and energy doing things that are often ineffective. When they fail to get results, they simply push harder, saying things like, "I can't stop now...I must keep going!"
- "Problem-solving kids" combine thinking and acting to solve problems effectively.
 - (i) They examine a problem to identify its root cause, develop a plan, take action, learn from both success and failures, and keep adjusting their plans as new challenges appear.

KEY QUOTES

"Problem solving isn't a talent that some people have and others don't. It's a habit. By developing the right skills and adopting the right attitude, anyone can become a problem-solving kid."

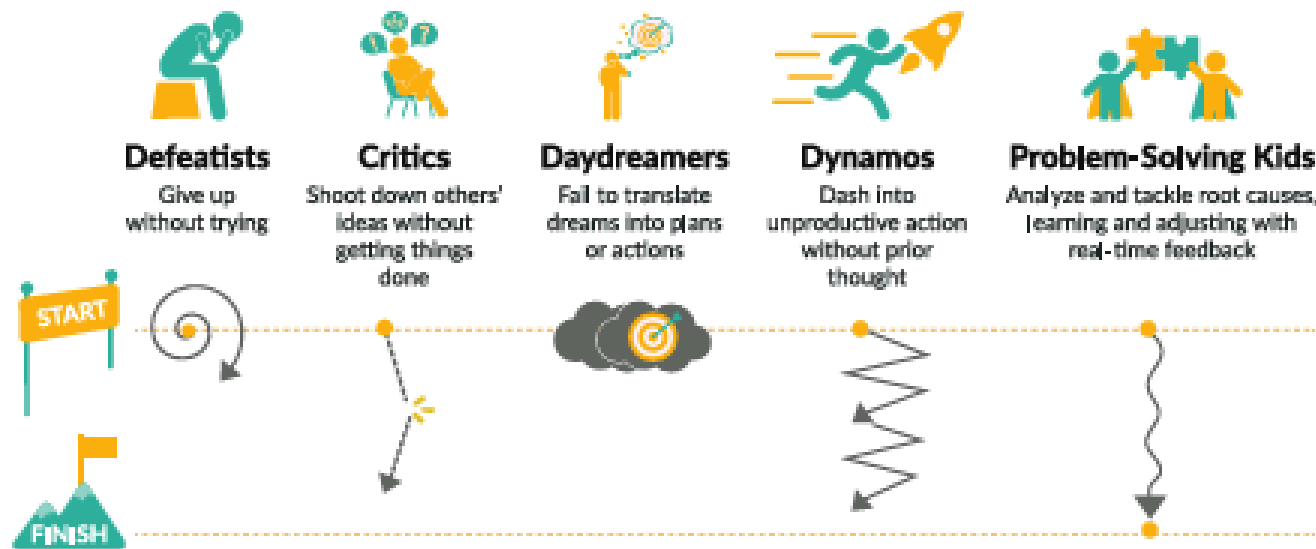
"Being a critic is easy; getting stuff done is the real challenge."

"Stopping to think can be just as important as taking action."

"Problem-solving kids have a real flair for setting goals and getting things accomplished. They take overcoming challenges in stride."

(ii) They break down ambitious goals and dreams into smaller, bite-sized milestones. They say things like, “What’s the root cause of this problem?” or “Let’s fix X, Y, and Z in 3 months.”

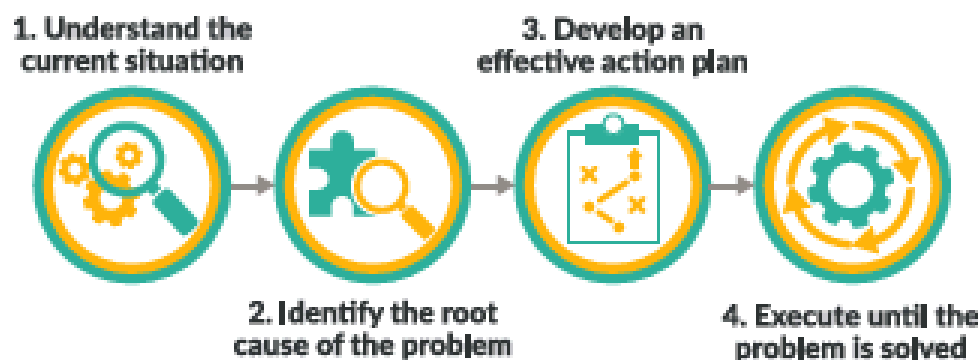
(iii) As a result, they achieve their goals more quickly, and also learn and evolve much faster than the other 4 types of problem-solvers.



The 4 Steps of Problem-Solving



When you see a doctor, they'd analyze your symptoms (e.g. fever, aches) to diagnose the root cause (e.g. flu), to prescribe the cure (flu medicine). Likewise, to solve any problem, you must first diagnose the root cause. Problem solving can be broken down into 4 steps:



- Understand the current situation or issue.
- Identify the root cause of the problem.
- Develop an effective action plan.
- Execute until the problem is solved, making adjustments as needed along the way.

KEY QUOTES

“Understand the situation, identify the root cause, develop an effective plan, and execute.”

 You need all 4 steps to solve a problem effectively. Once you've learned how to break a problem down into small, manageable chunks that you can solve, you can stay calm in face of any challenge—no matter how big or scary.

 Simple example: Let's imagine that you're trying to improve your physical fitness, but not getting the results you want. An average person might just assume that he/she must train harder. However, if you were a problem-solving kid, you'd approach it differently.

- Identify the problem: "I'm not seeing the improvement in physical fitness I want."
- Break down the problem into sub-parts. In this case, you might realize that you're falling short in 3 specific areas: cardiovascular endurance, muscle strength, and flexibility.
- From there, you'd analyze each area, then narrow down to potential root causes. For example:
 - (i) Cardio: Lack of intensity or variation;
 - (ii) Muscle strength: Incorrect weight or incomplete muscle group engagement; and
 - (iii) Flexibility: Consistently overlooking stretching.
- This level of specificity helps you to develop a targeted plan for each of the 3 areas of fitness, to deliver much better results than simply trying to "be more disciplined" or to "train more".

KEY QUOTES

"The value of problem-solving oriented thinking obviously extends far beyond the classroom into every facet of our lives. It enables us to take control of the challenges we encounter and to change the world in a positive way."



Part 2: Using the Problem-Solving Steps and Tools

Let's break down the 4 problem-solving steps into sub-parts, along with the tools you can use in each phase. Then, we'll illustrate them with 2 detailed examples.



Breakdown of the Problems-Solving Steps with Suggested Tools

 Step 1: Understand the situation so you can define the problem clearly and specifically.



Step 2: Identify the root cause of the problem. List down all potential root causes, and develop a hypothesis for the most likely root cause.

- List down all possible root causes, using a Logic Tree or a Yes/No Tree to systematically brainstorm possible causes.
- Develop a hypothesis for the most-likely root cause.
- Define the data/analyses needed to test the hypothesis. You can use a Hypothesis Pyramid to structure your assumptions and what you must do to validate them.
- Test the hypothesis to identify the root cause.

Key tools: Logic Tree, Yes/No Tree, Hypothesis Pyramid.



Step 3: Develop an effective action plan.

- Generate a wide range of solutions to the problem using the Logic Tree.
- Evaluate these options using the (i) Pros and Cons Grid and/or (ii) Criteria-Evaluation Table, and prioritize the best actions using the Prioritization Matrix.
- Organize these into an effective action plan.

Key tools: Logic Tree, Pros and Cons Grid, Criteria and Evaluation Table.



Step 4: Execute until the problem is solved, making adjustments as required. Monitor your progress and adjust the plan based on real-time feedback, learning from both successes and failures.



Your Problem-Solving Toolkit: Key Tools Explained



A logic tree helps you to (i) visualize the potential root causes of a problem, (ii) generate possible solutions systematically, and (iii) organize them into categories.

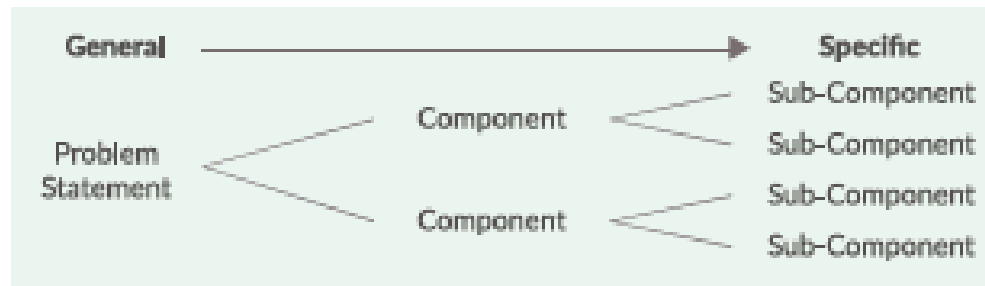
- State the main problem on the left.
- Then, branch out into possible solutions, ensuring no overlap and no omissions between categories. In other words:

KEY QUOTES

"The better you get at understanding the symptoms and identifying the root causes, the better you will get at developing effective solutions."

"Even if the problem you face is big and complicated, if you learn how to break it down into smaller, manageable problems, you will be able to solve it."

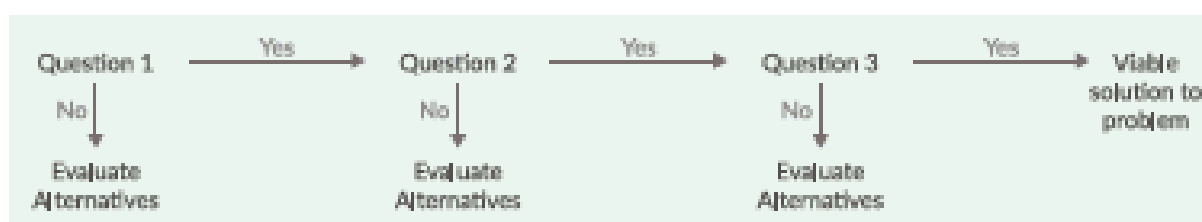
- (i) Make sure that each item in a category is mutually-exclusive (i.e. the items don't overlap); and
- (ii) The logic tree is comprehensive enough to cover all possible categories/aspects of a problem (i.e. nothing is left out).



- Expand the solutions in each branch or subcategory by asking: What? How? Where? When? Who?
- Try drawing or visualizing the logic tree to shift your perspective and deepen your understanding of the problem.
- Expand the tree horizontally and vertically:
 - (i) Expand vertically by asking if there are other ways of solving the problem.
 - (ii) Expand horizontally by asking if there are other aspects to the category.
- Keep adapting and adjusting your logic tree, organizing them from general (left) to specific (right), and grouping similar topics together.



A yes/no tree helps to ensure you've explored all potential solutions, and also guides decision-making. Keep asking yes/no questions until you find a viable solution.

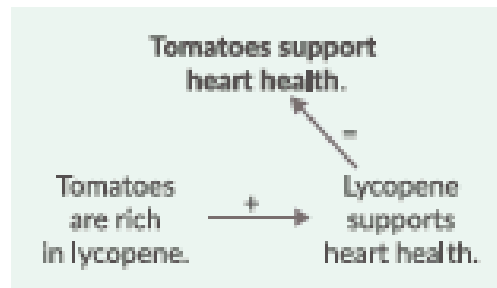
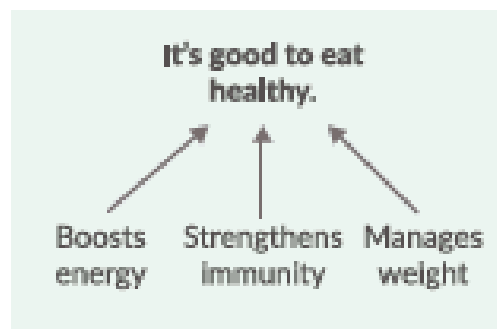


The clearer and more specific your hypotheses, the better you can test them to find an ideal solution. Here are several tools to help you structure your argument, and specify your rationale and conclusions, before you start to collect and analyze data. They also make it easier for you to communicate your hypothesis. You can:

KEY QUOTES

"You have to keep asking the "why" and the "how" to develop a custom-made action plan."

- Use a *hypothesis pyramid* by placing your hypothesis or conclusion at the top, with the supporting reasons below (why, what, where, how, who).
- Use a logic model (or an *argument structure*) to list down the statements that need to be true in order for the hypothesis or conclusion to be true.
- Create a *problem-solving design plan* to detail the problem to be solved, the hypothesis and rationale, and a list of actions and information needed to test the hypothesis. This is a dynamic document that evolves with each step of the problem-solving process.



Here are 2 additional tools to help you organize your information and thoughts. Remember: you can always brainstorm options to address specific challenges or shortcomings for each option.

- A *pros and cons grid* helps you to consider both the pros (positive aspects) and cons (negative aspects) of each option.

- List all the options.
- List the pros and cons for each option.
- Weigh each of the options.
- Select the best option.

Options	Pros	Cons
Option 1	+ Factor 1 + Factor 2	- Factor 1
Option 2	+ Factor 1	- Factor 1 - Factor 2 - Factor 3

- A *criteria-evaluation table* helps you to consider each option against your criteria and their relative importance.

- List all the options.
- List the evaluation criteria. Weigh each of the options.

Criteria	Importance of Criteria	Option 1	Option 2
Criteria 1	High	*****	***
Criteria 2	Med	*	*****

- Define the weightage for each criteria (e.g. Low/Medium/High).
- Evaluate each option based on the criteria (e.g. 1-5*).
- Choose the best option.

KEY QUOTES

"Challenge your own thinking processes and your conclusions."



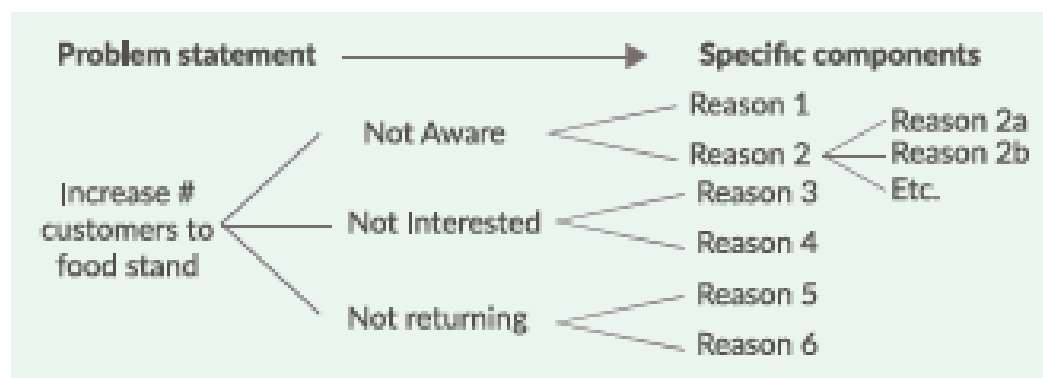
Example 1: Increasing Customers to a Pop-Up Food Stand

An enterprising student, Mimi, had a pop-up hot-dog stand in school and wanted to increase her number of customers.

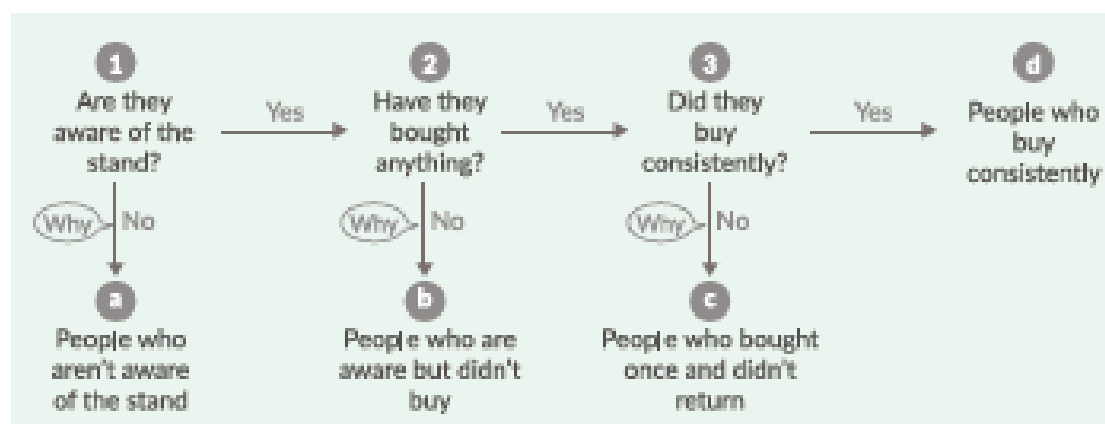


She started with Steps 1 and 2: Defining the problem and the possible root cause(s).

- Using a Logic Tree, she brainstormed potential causes for the low sales.
 - (i) Lack of awareness (people didn't know about the food stand).
 - (ii) Lack of interest (they were aware, but didn't buy).
 - (iii) They tried the food but didn't return.



- To ensure she has covered all scenarios, she also used a Yes/No Tree to explore the problem:



- Using the tools above, she brainstormed the possible reasons for the low sales at her food-stand, then made a best-guess of the most likely root cause. She hypothesized that the root cause was a lack of awareness. Specifically, she estimated that
 - (i) only 10% of the students knew about the stand, of which
 - (ii) 50% would buy and (iii) 80% would return.

KEY QUOTES

"Don't forget that the point of all this research is to help you make informed decisions."



Next, she defined the information needed to test her hypotheses, including the most efficient and effective way to collect the data.

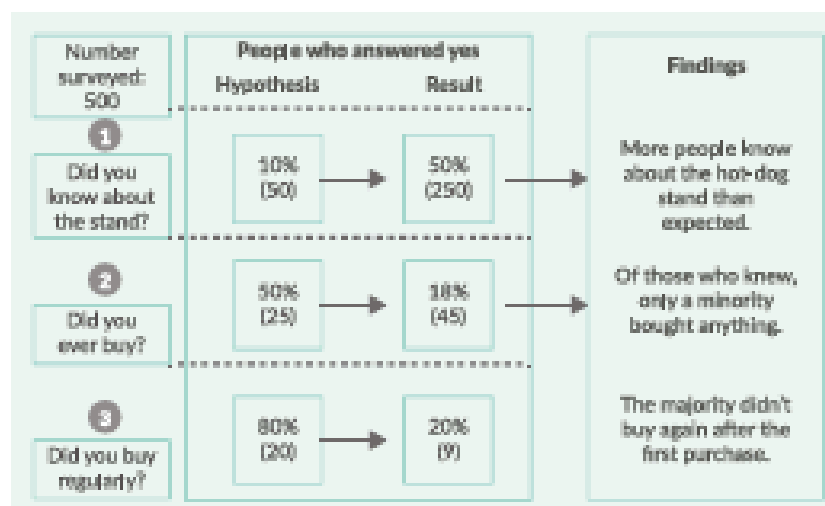
- She captured these in a problem-solving design plan:

Issue	Hypothesis	Rationale	Analysis/ Activities	Information Source
How many people are in each category?	Most people don't know about the stand	I haven't told many people about it.	Ask teachers to do a quick survey in class	Survey
Why are people who're aware not buying?	They don't like hot dogs	There's no other hot-dog stall in school.	Talk to 5 people who know but didn't buy	Interview
Will people want to keep buying?	Once they've tried, most will keep buying	Who doesn't like hot dogs? And my hot dogs are good!	Talk to 5 people who bought: will they buy again and why?	Interview

- For example, to find out the size of the 3 categories above, she asked her teachers for help, instead of doing individual surveys (which would be time-consuming). The teachers only had to ask 3 questions in class and count the number of students who raised their hands: Who knows about Mimi's food stand? Who has bought something from the stand? Who has bought something every week?
- To understand (a) why people who knew about her stall didn't buy, and (b) what she should do to get more regular customers, she asked the teachers to jot down several names of people in both groups so she could interview them later.



By testing her hypotheses, she realized that many more people knew about her stall (50%) than she expected, and she used interviews to find out why people weren't buying or returning.



KEY QUOTES

"You don't have to figure everything out on your own. Look for information to help you make the best decision in the given time."

KEY QUOTES

- People weren't buying mainly because (a) they didn't like hot dogs, or (b) her stall was only open on Friday after school, when the students wanted to hang out elsewhere.
- Those who tried her hot dogs but didn't buy again wanted more variety, such as different sauces or sides (e.g. fries or nuggets).



With the new insights, Mimi generated possible solutions for each of the 3 categories.

- Using the logic tree, she explored a wide range of solutions, then narrowed down to ideas that concurrently (i) increased awareness and (ii) made people want to buy. Here are 2 examples:

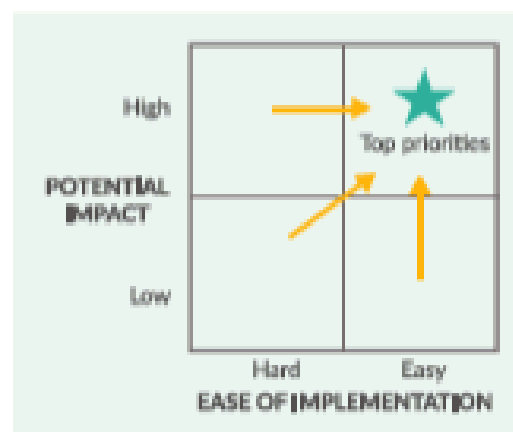
Who	Idea	Creates awareness?	Gets them to buy repeatedly?
Mimi	Give out flyers with promotions.	✓	✓
Customers	Bring a friend to get a free hot-dog.	✓	✓



Then, she developed an implementation plan.

- She prioritized the ideas based on impact and ease of implementation, in this sequence:

- Top-right box (high impact, easy to do);
- Top-left or bottom-right boxes;
- Bottom-left box (low impact, hard to do).



- With some brainstorming, she also managed to shift more items to the top-right box. For example, she didn't know how to design flyers, but found a friend who'd do it for her in exchange for free hot dogs.



Finally, she executed the plan, refining it as she went. Within the first 2 months, awareness increased to 80%, first-time

"You can accomplish more by collaborating with others who have strengths you may lack."

purchase increased to 50%, and repeat purchases to 40%. This meant a total of 200 customers each month, of which 80 were regular customers!



Example 2: Learning Roadmap for a Career Switcher

Alex was an experienced marketing manager who aspired to transition into data science. He broke down this ambition into smaller parts, and decided that his first step was to acquire the necessary skills through a reputable training program. To do so, he must choose an ideal course that fit with his budget and learning objectives.



Alex started by setting a clear, specific goal: to enroll in a part-time data science bootcamp within 6 months, ensuring the course (i) met his career needs, (ii) could be completed while holding his full-time job, and (iii) was within a budget of \$3,000.



Next, he assessed the gap between his goal and current situation, including:

- His current knowledge and skills, and the specific gaps in data science expertise he wished to fill;
- His available hours each week; and
- His financial situation: with estimated savings of \$300 per month, he needed an extra \$1,200 [$\$3,000 - (\$300 \times 6)$] for the course.



He shortlisted available courses, then used the following tools to compare and choose his ideal course:

- Pros and cons grid: to compare his top 5 options and narrow them down to 2.
- Criteria-evaluation table: He used this to compare the top 2 courses, taking into account factors like course content quality, flexibility, post-course support, and cost.

KEY QUOTES

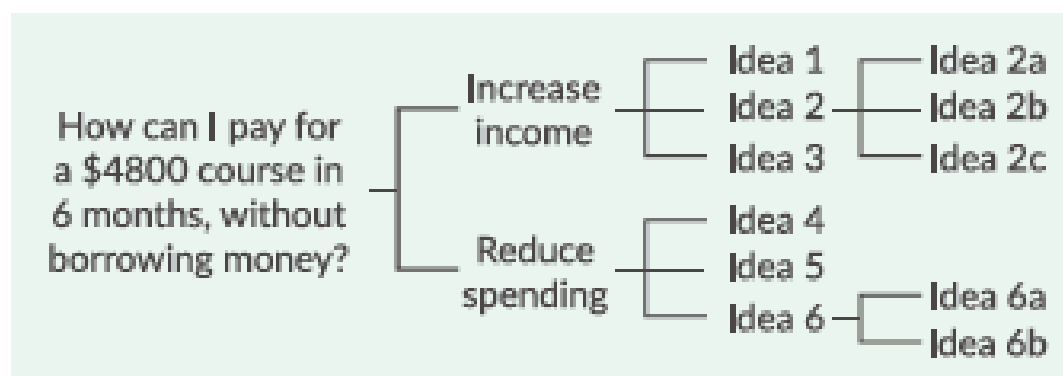
"The more specific the goal is, the more specific the action plan will be."

Criteria		Importance of criteria	Option 1	Option 2
Quality of Course	Covers all major topics?	High	All topics covered	All topics covered
	Has a mix of concepts + hands-on learning?	High	Mainly classroom learning.	Includes real-world projects
	Networking Opportunities?	Medium	Online forums	Alumni events & online community
Budget	Falls within \$3000?	High	\$2799	\$4800
Class schedule	Flexible learning hours?	Medium	Mainly online learning/ videos	Online learning + compulsory review & project sessions
	Less than 6 hours per week?	Low	Mainly lessons & quizzes	Extra simulations, presentations, and industry engagements

- Initially, Alex was leaning toward Option 1 as it covered all the essential topics, and fit both his budget and schedule. However, after doing further research and speaking with alumni and industry experts, he realized that the hands-on experience and networking opportunities from Option 2 would greatly benefit his career transition, even though it required more time and money.

 Using separate logic trees, Alex explored possible options/ideas to (i) meet his education budget, and (ii) free up more time without compromising his job.

- For example, he brainstormed ways to meet his education budget by (i) increasing his income (e.g. taking on side gigs), and (ii) reducing expenses (e.g. cutting down on meals at restaurants).



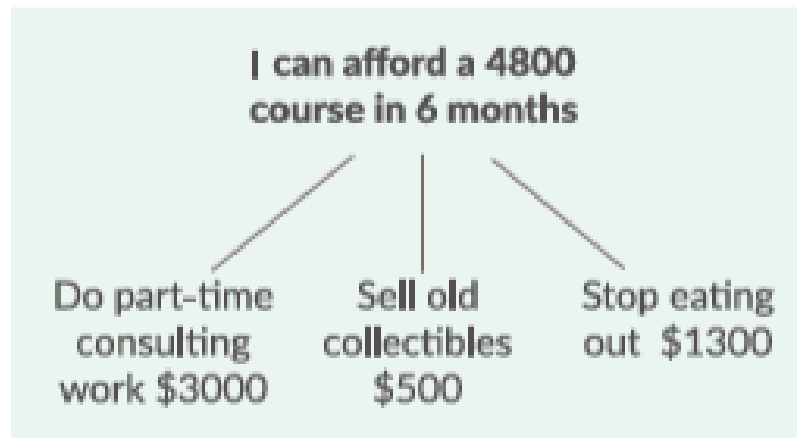
- He also did online research and spoke with friends, colleagues, and alumni members for ideas to effectively manage the part-time course while holding his full-time job.

KEY QUOTES

"Problem solving is easy when you know how to set a clear goal, figure out how to reach it, and follow through while reviewing your progress and making changes to your plan as necessary."



From there, he formed his hypothesis on the best ways to close the gap and achieve his goal. For example, he identified the top 3 ways to close the budget gap as: freelance consulting projects, selling old collectibles and reducing expenses on gadgets and restaurants.



Now, he had to test his hypotheses:

- Closing the budget gap:
 - (i) He researched the market rate and demand for freelance marketing consultants, and found it to be much more competitive than expected. He adjusted his projections to \$1000 instead of \$3000.
 - (ii) After putting his collectibles online, he was pleasantly surprised to receive quotes of up to \$2000.
 - (iii) After tracing his expenses, he realized he could save only \$500 from cooking instead of eating out, which might not be worth the hassle. Instead, he could save the same amount just by cutting back on beer alone.
 - (iv) In addition, he contacted the course organizers, and was excited to learn that he could qualify for an alumni-sponsored partial scholarship aimed at career-switchers like him, halving the cost to just \$2400!
- Additionally, Alex negotiated a flexible work arrangement with his employer for 3 days each week when he had to attend his course. He presented a case on how his training would bring data-driven decision-making skills to benefit his current role, and how he would complete his work remotely on work-from-home days.



With flexible work hours, a scholarship and additional income from the collectible sales, Alex only had to save an extra \$400

KEY QUOTES

*"Luck is what happens when preparation meets opportunity."
—Seneca*

"Spend less time worrying about things and more time thinking about actions you can take to get closer to your goals, then actually take action."

by reducing beer consumption. He was now ready to execute his final plan, enroll in the data science program, and start taking his first steps toward his dream career of a data scientist.

KEY QUOTES



Conclusion and Other Details in the Book



As we saw in the 2 examples above, you can solve problems and achieve your goals more effectively by researching the problem, exploring the issues and solutions, planning and testing your hypotheses, asking for help, and constantly challenging your own thought processes and conclusions.



The book is full of illustrations, charts, tables, and examples to bring the ideas, steps and tools to life. Specifically, Ken Watanabe presents 3 detailed case studies of:

- The Mushroom Lovers: A new rock band that wanted to improve their concert attendance;
- John Octopus: A young man who aspired to become a computer graphics animator and Hollywood movie director, and needed to buy a computer to start learning CGI; and
- Kiwi: a soccer player who wanted find the best school in Brazil to hone her soccer skills and become bilingual/bicultural.

"If you make problem solving a habit, you'll be able to make the most of your talents and take control of your life. You can solve not only your own problems, but the problems of your school, your business, and your community—and maybe even the world."



About the Author

Ken Watanabe grew up bilingual in Japan and studied in the United States at Yale and Harvard Business School. He was a management consultant at McKinsey & Company for six years. He is now the founder and CEO of his own education, entertainment and media company, Delta Studio.

